

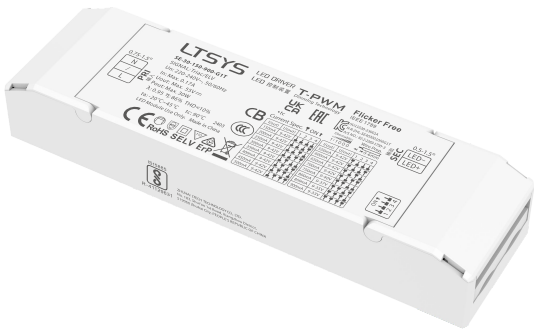
Intelligent LED Driver (Constant Current)

- The housing is made from V0 flame retardant PC materials from SAMSUNG/COVESTRO.
- Small size and light weight. The clamshell design and screwless type for strain-relief.
- Support Leading edge(Triac), Trailing edge(ELV).
- With soft-on and fade-in dimming function, enhancing your visual comfort.
- T-PWM™ dimming technology allows continuous and flicker-free images under high-speed photography.
- Dimming from 0~100%, down to 0.1%.
- The whole dimming process is flicker-free with high frequency exemption level.
- Multiple current levels and wide voltage range. Suitable for different power of LEDs.
- Class 2 LED driver, Safety Extra Low Voltage (SELV).
- Innovative thermal management technology intelligently protects the life of the LED driver.
- Overheat, overload, short circuit protection and automatic recovery.
- Suitable for Class I / II / III indoor light fixtures.
- Up to 50,000-hour life time.
- 5-year warranty (Rubycon capacitor).

T-PWM
Dimming Technology

Flicker Free
IEEE 1789

Dimmable:
1:1000



Triac
ELV

T-PWM™
Dimming Technology

Flicker Free
IEEE 1789

ON
1 2 3 4
Multiple
Current

Short Circuit
Protection

Overload
Protection

Overheat
Protection

Technical Specs

Model		SE-40-300-1050-G1T		SE-30-150-900-G1T	
Features	Output Type	Constant Current			
	Dimming Interface	Triac/ELV			
	Output Feature	Isolation			
	Protection Grade	IP20			
	Insulation Grade	Class II (Suitable for class I/ II /III light fixtures)			
OUTPUT	Output Voltage	9-42Vdc			
	Maximum output voltage	≤55V			
	Output Current Range	300-1050mA		150-900mA	
	Output Power Range	2.7W-40W		1.35W-30W	
	Dimming Range	0-100%, down to 0.1%			
	LF Current Ripple(<120Hz)	<3%			
	Current Accuracy	±5%			
	Ripple & Noise	≤5V			
PWM Frequency	3600Hz				
INPUT	DC Voltage Range	200-280Vdc			
	Input Voltage	220-240Vac			
	Frequency	50/60Hz			
	Input Current	≤0.22A/230Vac		≤0.17A/230Vac	
	Power Factor	PF>0.95/230Vac, at full load			
	THD	THD<10%/230Vac, at full load			
	Efficiency (Typ.)	>88%@950mA		>86%@750mA	
	Inrush Current	Cold start 16A[Test twidth=90us tested under 50% Ipeak]/230Vac			
Anti Surge	L-N: 2kV				
Leakage Current	<0.5mA/230Vac				
ENVIRONMENT	Working Temperature	ta: -20 ~ 45°C tc: 90°C			
	Working Humidity	20 ~ 95%RH, non-condensing			
	Storage Temperature/Humidity	-40 ~ 80°C/10-95%RH			
	Temperature Coefficient	±0.03%/°C (-20~45°C)			
	Vibration	10-500Hz, 2G 12min/1cycle, 72 min for X, Y and Z axes respectively			
PROTECTION	Overload Protection	Shut down the output and recover automatically once it exceeds 1.02-1.35 times of the rated power			
	Overheat Protection	Intelligently adjust or turn off the current output if the PCB temperature ≥110°C. When the PCB temperature <90°C, automatically recover normal output			
	Short Circuit Protection	When short circuit occurs, shut down the output and recover automatically			
SAFETY & EMC	Withstand Voltage	I/P-O/P: 3750Vac			
	Insulation Resistance	I/P-O/P: 100MΩ/500VDC/25°C/70%RH			
	Safety Standards	CCC	China	GB19510.1, GB19510.14	
		TUV	Germany	EN61347-1, EN61347-2-13, EN62493	
		CE	European Union	EN61347-1, EN61347-2-13, EN62384	
		KC	Korea	KC61347-1, KC61347-2-13	
		RCM	Australia	AS61347-1, AS61347-2-13	
		ENEC	Europe	EN61347-1, EN61347-2-13, EN62384	
		CB	CB Member States	IEC61347-1, IEC61347-2-13	
		EAC	Russia	IEC61347-1, IEC61347-2-13	
	BIS	India	IS 15885(PART 2/SEC 13)		
	EMC Emission	CCC	China	GB/T17743, GB17625.1	
		CE	European Union	EN55015, EN61000-3-2, EN61000-3-3, En61547	
		KC	Korea	KN15, KN61547	
		RCM	Australia	EN55015, EN61000-3-2, EN61000-3-3, EN61547	
EAC		Russia	IEC 62493, IEC 61547, EH 55015, IEC 61000-3-2, IEC 61000-3-3		
EMC Immunity	EN6100-4-2,3,4,5,6,8,11, EN61547				
ErP	Power Consumption	Standby power consumption	No standby mode		
		Networked standby	No networked standby mode (No Phase-cut signal, no power consumption)		
		No-load power consumption	Without no-load mode		
	Flicker/Stroboscopic Effect	IEEE 1789	Meet IEEE 1789 standard/High frequency exemption level		
		CIE SVM	Pst LM≤1.0, SVM≤0.4		
OTHERS	DF	Phase factor	DF≥0.9		
	Weight(N.W.)	163g±10g			
	Dimensions	142×40×23mm(L×W×H)			

LED Current Selection



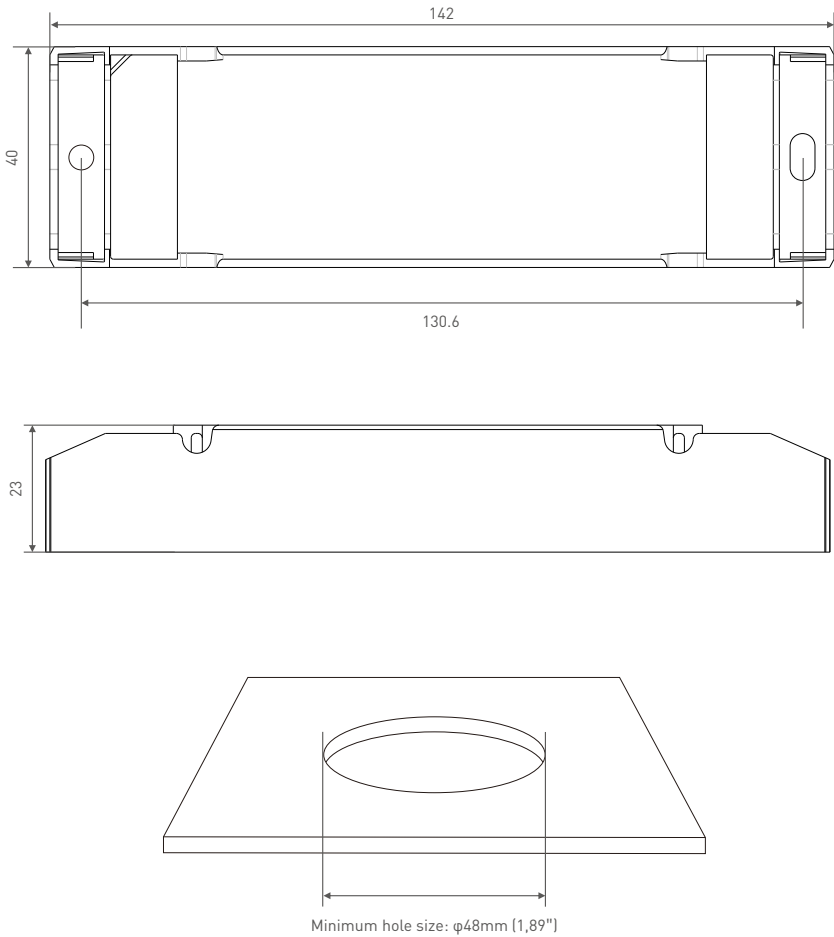
SE-40-300-1050-G1T	DIP Switch	⬇⬇⬇⬇	⬇⬇⬇⬆	⬇⬇⬆⬇	⬇⬇⬆⬆	⬇⬆⬇⬇	⬆⬆⬆⬆	⬆⬆⬆⬇	⬆⬆⬆⬆	ON OFF
	Output Current	300mA	350mA	400mA	450mA	500mA	550mA	600mA	650mA	
	Output Voltage	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	
	Output Power	2.7-12.6W	3.15-14.7W	3.6-16.8W	4.05-18.9W	4.5-21W	4.95-23.1W	5.4-25.2W	5.85-27.3W	
	DIP Switch	⬆⬇⬇⬇	⬆⬇⬇⬆	⬆⬇⬆⬇	⬆⬇⬆⬆	⬆⬆⬇⬇	⬆⬆⬇⬆	⬆⬆⬆⬇	⬆⬆⬆⬆	
	Output Current	700mA	750mA	800mA	850mA	900mA	950mA	1000mA	1050mA	
	Output Voltage	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	9-40V	9-38V	
SE-30-150-900-G1T	DIP Switch	⬇⬇⬇⬇	⬇⬇⬇⬆	⬇⬇⬆⬇	⬇⬇⬆⬆	⬇⬆⬇⬇	⬆⬆⬆⬆	⬆⬆⬆⬇	⬆⬆⬆⬆	ON OFF
	Output Current	150mA	200mA	250mA	300mA	350mA	400mA	450mA	500mA	
	Output Voltage	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	9-42V	
	Output Power	1.35-6.3W	1.8-8.4W	2.25-10.5W	2.7-12.6W	3.15-14.7W	3.6-16.8W	4.05-18.9W	4.5-21W	
	DIP Switch	⬆⬇⬇⬇	⬆⬇⬇⬆	⬆⬇⬆⬇	⬆⬇⬆⬆	⬆⬆⬇⬇	⬆⬆⬇⬆	⬆⬆⬆⬇	⬆⬆⬆⬆	
	Output Current	550mA	600mA	650mA	700mA	750mA	800mA	850mA	900mA	
	Output Voltage	9-42V	9-42V	9-42V	9-42V	9-40V	9-37V	9-35V	9-33V	
	Output Power	4.95-23.1W	5.4-25.2W	5.85-27.3W	6.3-29.4W	6.75-30W	7.2-29.6W	7.65-29.75W	8.1-29.7W	

* After setting the current via DIP switches, power off and then power on the driver to make the new current setting effective.

* E.g. LED 3V/pcs: 9-42V can power 3-14pcs LEDs in series, 9-21.5V can power 3-7pcs LEDs, the max quantity of LEDs in series will be subject to the actual voltage of LEDs.

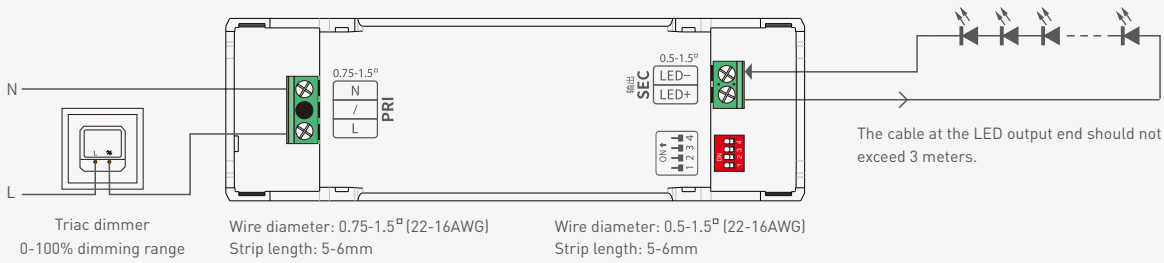
Product Size

Unit: mm

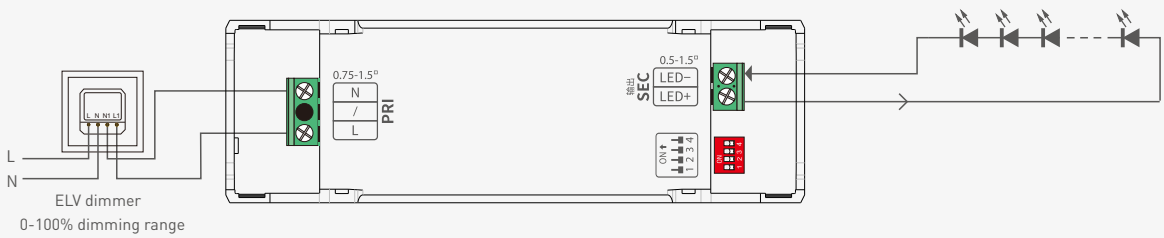


Wiring Diagram

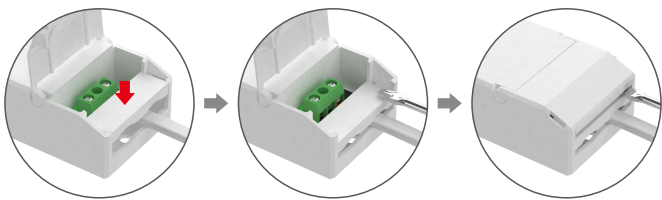
Triac Connection



ELV Connection



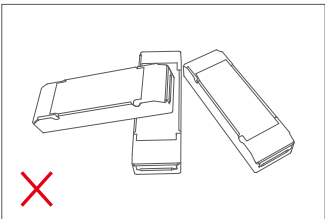
Application Diagram of Protective Cover



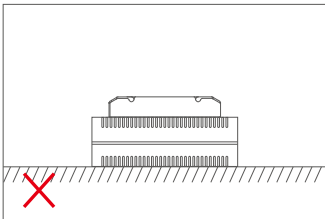
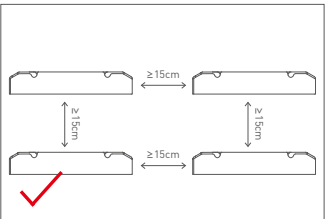
1. Put the head of a screwdriver on the side of the housing to pry up both the protective cover and wire fixing board. Then remove the wire fixing board and connect to the wires as wiring diagram shows.

2. Install the wire fixing board and press it down. Then snap on the protective cover while pressing the wire fixing board with a small flat-head screwdriver

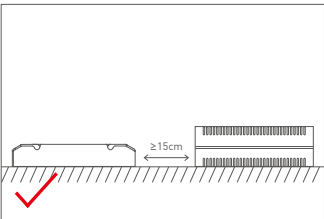
Installation Precautions



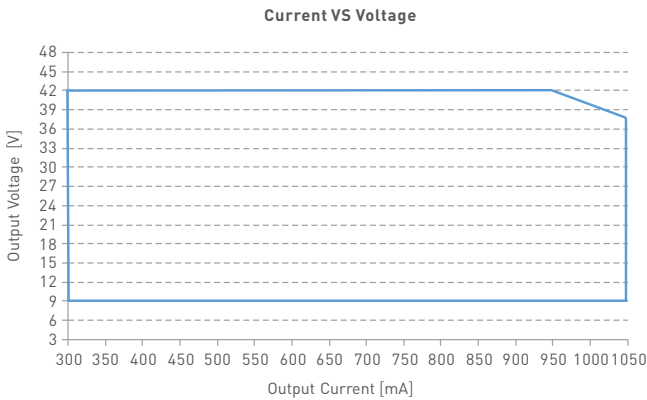
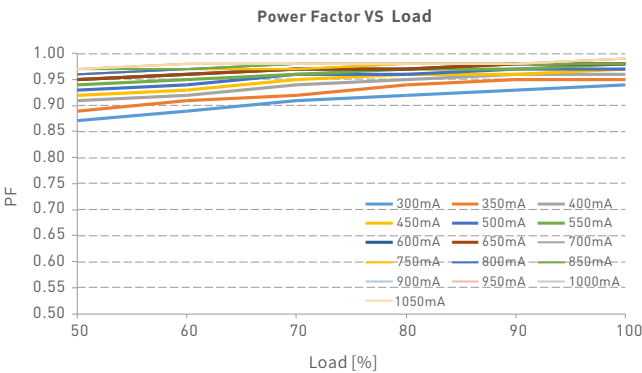
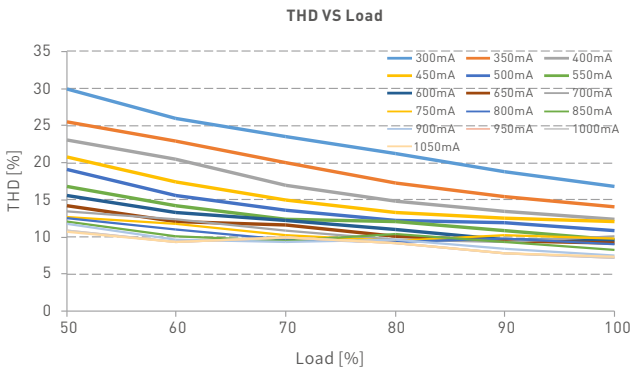
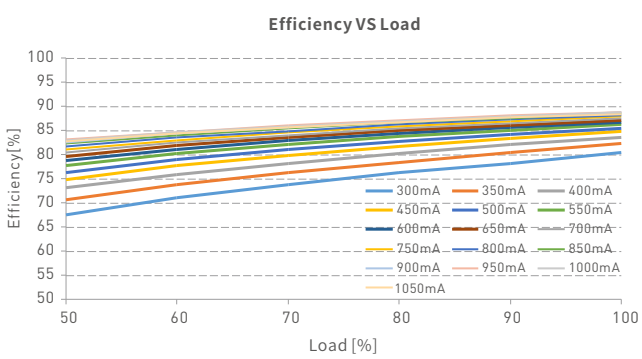
Please do not stack the products. The distance between two products should be ≥15cm so as not to affect heat dissipation and the lifespan of the products.



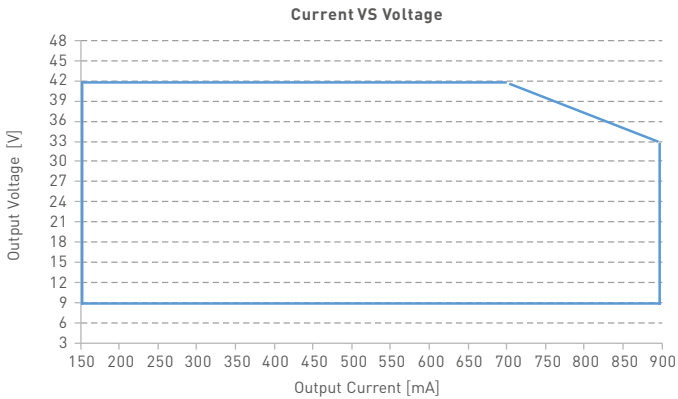
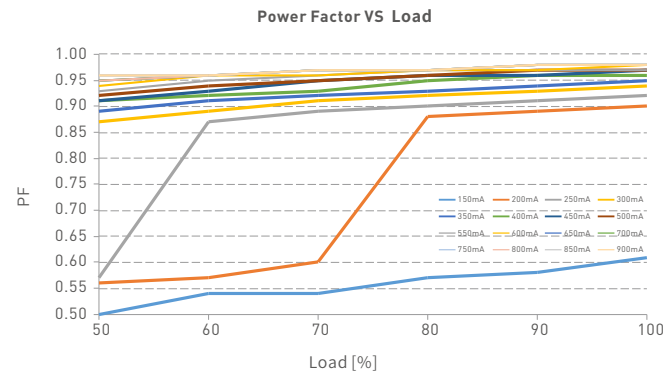
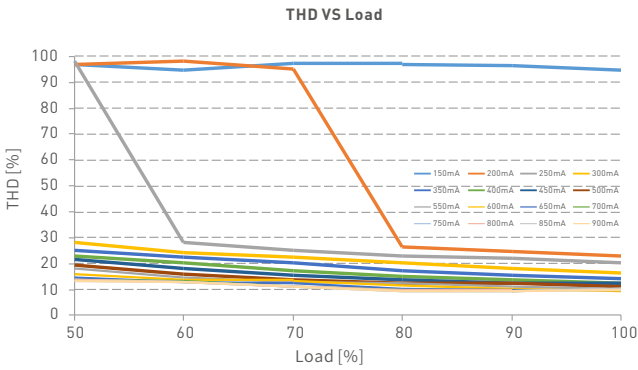
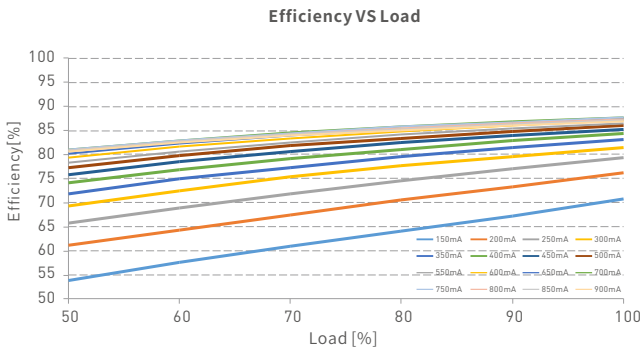
Please not place the products on LED drivers. The distance between the product and the driver should be ≥15cm so as not to affect heat dissipation and shorten the lifespan of the products.



Relationship Diagrams



SE-40-300-1050-G1T



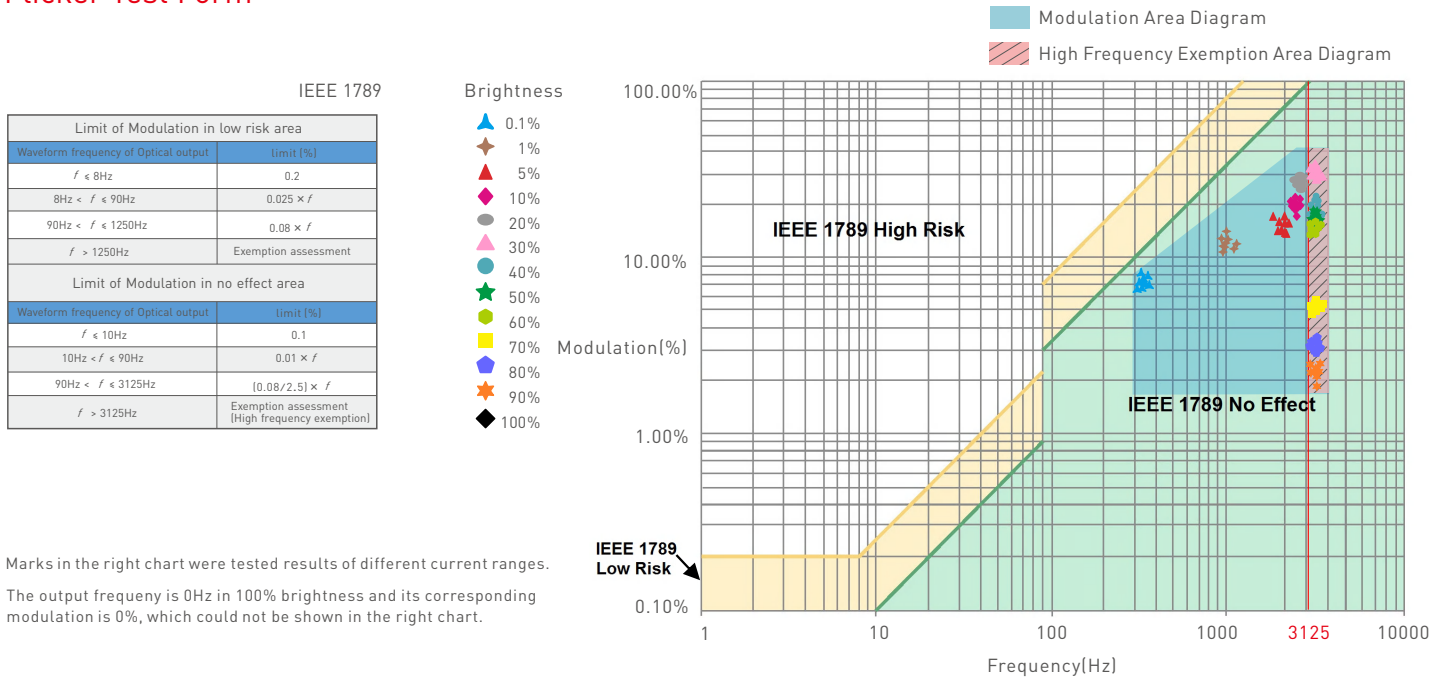
SE-30-150-900-G1T

Surge Current & Corresponding Miniature Circuit Breaker (MCB) Load Capacity Table

MCB Model	B10	B13	B16	B20	B25	C10	C13	C16	C20	C25	D10	D13	D16	D20	D25
Maximum Load Capacity	15	19	24	30	40	17	22	27	35	43	20	25	31	39	49

- Remarks
1. Test Conditions Cold start 16A[Test twidth=90us tested under 50% Ipeak]/230V~
 2. The number of supported drivers may vary depending on the brand and model of the MCB.
 - 3.It is recommended not to exceed the specified load capacity during on-site installation. The actual load should be determined based on field conditions.
 - 4.If the ambient temperature exceeds 30°C or multiple MCBs are installed side by side, the number of installed drivers must be reduced and recalculated accordingly.
 - 5.Electricians typically use Type B MCBs for residential lighting and Type C MCBs for commercial lighting applications.
 - 6.Different testing equipment may yield variations in measured current peaks and pulse widths. Always use professional-grade instruments for accurate testing.

Flicker Test Form



Packaging Specifications

Model	SE-40-300-1050-G1T / SE-30-150-900-G1T
Carton Dimensions	320×275×106mm(L×W×H)
Quantity	20 PCS/Layer; 2 Layers/Carton; 40 PCS/Carton
Weight	0.163 kg/PC; 7.32 kg/Carton

Packaging Image



Inner Packaging Box

Carton Packaging

Transportation and Storage

1. Transportation

Products can be shipped via vehicles, boats and planes.

During transportation, products should be protected from rain and sun. Please avoid severe shock and vibration during the loading and unloading process.

2. Storage

The storage conditions should comply with the Class I Environmental Standards. The products that have been stored for more than six months are recommended to be re-inspected and can be used only after they have been qualified.

Attentions

- This product must be installed and adjusted by a qualified professional.
- This product is non-waterproof (special models excepted). Please avoid the sun and rain. When installed outdoors, please ensure it is mounted in a water proof enclosure.
- Good heat dissipation will extend the life the product. Please install the product in a environment with good ventilation.
- When you install this product, please avoid being near a large area of metal objects or stacking them to prevent signal interference.
- Please keep the product away from a intense magnetic field, a high pressure area or a place where lightning is easy to occur.
- Please check whether the working voltage used complies with the parameter requirements of the product.
- Before you power on the product, please make sure all the wiring is correct in case of incorrect connection that may cause a short circuit and damage the components, or trigger a accident.
- If a fault occurs, please do not attempt to fix the product by yourself. If you have any question, please contact the supplier.

* This manual is subject to changes without further notice. Product functions depend on the goods. Please feel free to contact our official distributors if you have any question.

Warranty Agreement

- Warranty periods from the date of delivery: 5 years.
- Free repair or replacement services for quality problems are provided within warranty periods.

Warranty exclusions below:

- Beyond warranty periods.
- Any artificial damage caused by high voltage, overload, or improper operations.
- Products with severe physical damage.
- Damage caused by natural disasters and force majeure.
- Warranty labels and barcodes have been damaged.
- No any contract signed by LTECH.

1. Repair or replacement provided is the only remedy for customers. LTECH is not liable for any incidental or consequential damage unless it is within the law.
2. LTECH has the right to amend or adjust the terms of this warranty, and release in written form shall prevail.